

The schematic diagram illustrates the electrical connections for a USB Type-C connector and a DC-CC Buck converter.

USB Type-C Connector (Left):

- Connector:** TYPE C-DB-116PWB
- Pin 1 (VBUS):** Connected to the VBUS supply.
- Pin 2 (D+):** Connected to the D+ signal line.
- Pin 3 (D-):** Connected to the D- signal line.
- Pin 4 (GND):** Connected to ground.
- Pin 5 (GND):** Connected to ground.
- Pin 6 (GND):** Connected to ground.
- Pin 7 (GND):** Connected to ground.
- Pin 8 (GND):** Connected to ground.
- Pin 9 (GND):** Connected to ground.
- Pin 10 (GND):** Connected to ground.
- Pin 11 (GND):** Connected to ground.
- Pin 12 (GND):** Connected to ground.
- Pin 13 (GND):** Connected to ground.
- Pin 14 (GND):** Connected to ground.
- Pin 15 (GND):** Connected to ground.
- Pin 16 (GND):** Connected to ground.
- Pin 17 (GND):** Connected to ground.
- Pin 18 (GND):** Connected to ground.
- Pin 19 (GND):** Connected to ground.
- Pin 20 (GND):** Connected to ground.

DC-DC Buck Converter (Right):

- IC:** TPS62840YBGR
- Input:** VBUS (5V) connected to VIN (pin 1).
- Output:** 3.3V connected to VOUT (pin 2).
- Grounding:** GND connected to GND (pin 3).
- Capacitors:** C6 (4.7 uF) at VIN, C9 (10 uF) at VOUT.
- Inductor:** L2 (2.2 uH) between VIN and VOUT.
- Control:** VSET (pin 4) connected to GND, SW (pin 5) connected to GND, EN (pin 6) connected to GND.
- Load:** R8 (180 kohm) connected to VOUT and GND.

The diagram illustrates an I2C BUS ACCELERATOR circuit. It features two sensors, SENSOR1 and SENSOR2, connected to an I2C bus. The circuit includes a cable connector (CON) with pins 1 through 7, connected to a 3.3V supply and ground. The I2C bus is connected to the SCL and SDA lines of both sensors. The I2C bus accelerator chip (U17) is connected to the 3.3V supply and ground, with its ENABLE pin connected to the SCL line. The address settings for the sensors are 0X77 for SENSOR1 and 0X76 for SENSOR2.

SENSOR1
 U19: MS560702BA03-50
 ADDRESS SETTING: 0X77
 Pins: 1 VDD, 2 PS, 3 GND, 4 CSB, 5 CSB, 6 SDO, 7 SCL, 8 SCL/SDA

SENSOR2
 U18: BME680
 ADDRESS SETTING: 0X76
 Pins: 1 GND, 2 CSB, 3 VDDIO, 4 SCK, 5 SDO, 6 VDDIO, 7 GND, 8 VDD

I2C BUS ACCELERATOR
 U17: LTC4311IDC#TRPBF
 Pins: 1 ENABLE, 2 NC, 3 VCC, 4 GND, 5 BUS2, 6 GND, 7 EP

CABLE CONNECTOR
 CON: 1 3.3V, 2 GND, 3 SCL, 4 GND, 5 SDA, 6 GND, 7 SHIELD

400 KHZ , 2M CABLE

Normal Voltage Mode